

Marketing Campaign Analysis — Phase 5: Sentiment Analysis

Omar Touzani · 25,668 Amazon reviews · Goal: identify which model best catches customer complaints

I trained three text classifiers on TF-IDF features (unigrams + bigrams) to predict sentiment from review text. The dataset is ~90% Positive, so model selection is driven by Negative-class recall — the fraction of actual complaints the model catches. Accuracy on a 90/5/5 imbalance is dominated by the Positive majority and doesn't measure the thing the business needs.

Model comparison — confusion matrix → report → AUC-ROC

Model	Accuracy	AUC-ROC	Neg recall	Neu recall	Pos recall	Decision
Naive Bayes	0.9289	0.9341	0.5475	0.1498	0.9926	Misses 45% of complaints
Logistic Regression	0.8880	0.9310	0.8000	0.6916	0.9035	Selected — best complaint catcher
Linear SVC	0.9414	0.9334	0.6557	0.3392	0.9900	Highest accuracy — but misses 34% of complaints

Why Logistic Regression over Linear SVC. Linear SVC has the highest accuracy (94.1%), but it only catches 66% of complaints. Logistic Regression catches 80% — that's 44 more real complaints identified out of 305 in the test set, a 21% improvement on the metric that matches the business question. The accuracy gap (88.8% vs 94.1%) is the price paid for that improvement, and it comes almost entirely from misclassifying some Positive reviews as Neutral — a much cheaper error than missing complaints.

Dominant complaint theme — battery life

The top Negative TF-IDF words are not scattered — they tell one clear story. The Negative class is dominated by: **battery, last, long, dead, "don't last", week, return, disappoint**. Customers complaining about Amazon products are overwhelmingly complaining about battery life — not delivery, not packaging, not price. If this dataset's products were under the team's control, battery durability would be the single highest-impact product-quality fix.

Top categories by complaint rate

Category	Total reviews	Negative	Negative %	Avg rating
Health & Beauty	9,465	1,090	11.52%	4.37
Toys & Games, Electronics	1,663	54	3.25%	4.53
Electronics	13,955	370	2.65%	4.56
Electronics, Media	185	4	2.16%	4.66
Office Supplies, Electronics	386	7	1.81%	4.55

Health & Beauty stands alone — 11.5% negative rate is 4× the Electronics rate. Combined with the battery theme, the implication is that battery-powered Health & Beauty products (electric toothbrushes, shavers, scales, beauty devices) are the highest-complaint pocket of the catalogue.

Caveats

Neutral is genuinely hard to classify. Neutral reviews share most of their vocabulary with Negative reviews (battery, last, don't, work) — they're essentially "concerns without anger". Even the selected model only catches 69% of Neutral reviews. For a complaint-detection use case this is fine; the few Neutral reviews predicted as Negative are still actionable feedback.

Standalone scope. Reviews and campaign customers share no join key. These insights are about reviewer sentiment in general, not about how Marketing Intelligence Platform campaign customers feel.

What's next

Phase 6 — Market basket analysis on the cleaned campaign spending data. Identify which product categories are purchased together and how patterns differ by CLV tier from Phase 4.